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Language Ranker: A Metric for Quantifying LLM Performance Across High and Low-Resource Languages

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Abstract

The development of Large Language Models (LLMs) relies on extensive text corpora, which are often unevenly distributed across languages. This imbalance results in LLMs performing significantly better on high-resource languages like English, German, and French, while their capabilities in low-resource languages remain inadequate. Currently, there is a lack of quantitative methods to evaluate the performance of LLMs in these low-resource languages. To address this gap, we propose the *Language Ranker*, an intrinsic metric designed to benchmark and rank languages based on LLM performance using internal representations. By comparing the LLM's internal representation of various languages against a baseline derived from English, we can assess the model's multilingual capabilities in a robust and language-agnostic manner. Our analysis reveals that high-resource languages exhibit higher similarity scores with English, demonstrating superior performance, while low-resource languages show lower similarity scores, underscoring the effectiveness of our metric in assessing language-specific capabilities. Besides, the experiments show that there is a strong correlation between the LLM's performance in different languages and the proportion of those languages in its pre-training corpus. These insights underscore the efficacy of the Language Ranker as a tool for evaluating LLM performance across different languages, particularly those with limited resources.

Introduction

Large Language Models (LLMs), such as GPT-4, Claude-3 and LlaMa-3, have demonstrated remarkable performance in various NLP tasks (Achiam et al. 2023; Ouyang et al. 2022; Touvron et al. 2023; Team et al. 2024; Jiang et al. 2023; Bai et al. 2023). However, this progress masks a critical issue: the stark disparity in LLM performance across languages, particularly disadvantaging low-resource languages common in developing countries. This disparity stems from the overwhelming bias towards high-resource languages, especially English, in training datasets (Xie et al. 2024). For instance, approximately 92.65% of GPT-3's training tokens are in English, leaving a mere 7.35% for all other languages (OpenAI 2023). Similarly, English accounts for 89.70% of LlaMa2's pre-training data (Touvron et al. 2023).

This imbalance leads to significant performance gaps between high-resource languages (e.g., English, German, French) and low-resource languages, potentially exacerbating global digital divides. LLMs often struggle with context-specific interpretations in low-resource languages, such as understanding culturally specific expressions or idioms (Zhang et al. 2023). Recent studies confirm that pre-trained models perform poorly in languages with insufficient training data (Lankford, Afli, and Way 2024), highlighting the urgent need for robust methods to quantify and address these disparities. It is urgent to develop a tool to quantify linguistic biases in LLMs, so as to contribute to more equitable AI technologies, potentially improving access and performance for underserved language communities worldwide.

To tackle this challenge, we introduce *Language Ranker*, a novel metric designed to systematically evaluate LLM performance across diverse languages, with a particular focus on low-resource languages. Our method leverages internal representations of LLMs, establishing English corpus representations as a baseline and measuring the similarity between this baseline and representations from other languages. This similarity score serves as a quantitative measure of language-specific performance. We validate Language Ranker by applying it to five state-of-the-art LLMs: LlaMa2 (Touvron et al. 2023), LlaMa3 (Meta-AI 2024), Qwen (Bai et al. 2023), Mistral-v0.1 (Jiang et al. 2023), and Gemma (Team et al. 2024). Additionally, we analyze the relationship between Language Ranker scores, the proportion of languages in training datasets, and performance on established benchmarks. Our comprehensive experiments yield the following key findings:

- Experimental results indicate that high-resource languages consistently show higher similarity scores with English, while low-resource languages exhibit lower scores, validating Language Ranker's effectiveness in quantifying language-specific performance disparities.
- We uncover a strong correlation between LLM performance and the proportion of languages in pre-training corpora, providing crucial insights into the impact of training data distribution on multilingual capabilities.
- The analysis of embedding space distributions reveals that high-resource languages are more evenly distributed, while low-resource languages cluster narrowly, further supporting the reliability of the proposed metric.

* Work done during Zihao Li's remote internship at NJIT.

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The Proposed Method

In this section, we will give an introduction to our analysis method. First, we will introduce the dataset that we used. Then, we will introduce how to obtain the similarity between English and other languages, as well as how to compare different LLMs' performances. [1]

Probing Datasets

We use OPUS-100 (Zhang et al. 2020) as our evaluation datasets. OPUS-100 is an English-centric multilingual corpus that covers 100 languages. Each sample consists of text in a non-English language as the original data, with its English translation serving as the target data. For example, {"German": "Ich wollte dir erst noch etwas zeigen.","English": "I wanted to show you something first."}. After filtering, there are 94 subsets containing English, including high-resource languages such as German, French, and Chinese, as well as low-resource languages such as Oriya, Kannada, and Kazakh. Each subset contains 2000 samples.

Similarity Measurement

We employ cosine similarity to measure the LLMs' performance gap between the target language and English. Specifically, given two sentences $X = \{x_i\}_{i=1}^{n}$ and $Y = \{y_i\}_{i=1}^{m}$ representing the text in English and the text in the target language. Since the current LLMs are all autoregressive models, we use the representation obtained after LLM mapping of the last token x_n and y_m as the representation of the text and calculate the similarity between them. As we know, LLM consists of several layers of Transformer blocks (Vaswani et al. 2017). Therefore, after each layer of mapping by the transformer block, we can get a representation vector x_n^l and y_m^l, $l = 1...H$, where H represents the number of the layer of LLMs. According to (Li et al. 2024), the intermediate representation can be briefly summarized by the following equations:

$$x^{l+1} = \text{MLP}(x^l + \text{MHA}(x^l)) \quad l = 1...H, \tag{1}$$

where MHA means multi-head attention or multi-group attention, and MLP means standard multilayer perceptron layer. Next, we take x_n^l and y_m^l to calculate the similarity. To implement a more robust similarity measure, we use the average similarity obtained by several intermediate layers as the final similarity. This process can be described as follows:

$$Sim = \frac{1}{|l_{sub}|} \sum_{i=1}^{|l_{sub}|} Sim_i, \text{ where } Sim_i = \frac{x_n^i y_m^i}{||x_n^i|| ||y_m^i||}, \tag{2}$$

where l_{sub} = $\{5, 10, 15, 20, 25\}$ is the subset of the layers we selected. Finally, we use Sim to evaluate the performance gap between English and Non-English corpus.

Rank Correlation Measurement

When we get the similarity between each non-English representation and the English representation, we sort them according to the similarity to get a sorted ranking list of all languages. To measure the similarity of the sorted ranking lists of two LLMs, we use the longest common partial order sublist to measure. It can be defined as follows: For two sorted lists A and B, find a sublist C that is a subset of A and B such that for any number of index $i_1 \leq i_2 \leq ... \leq i_n$, Index($C_{i_1}$)≤Index($C_{i_2}$)≤...≤Index($C_{i_n}$) is true for both A and B, and the longest sublist C that makes it true is called the longest common partial order sublist of A and B. We use the ratio of the length of the longest common partial order sublist of two LLMs to the total length of the ranking list as a metric to measure the correlation.

Experiments

In our experiments, we utilize five prominent open-source large models: LlaMa2 (Touvron et al. 2023), LlaMa3 (Meta-AI 2024), Qwen (Bai et al. 2023), Mistral-v0.1 (Jiang et al. 2023), and Gemma (Team et al. 2024).

We conduct experiments to answer the following research questions in the following four subsections respectively: RQ1: Can the Language Ranker effectively quantify the performance of LLMs across multiple languages? RQ2: How consistent are the performance rankings of different LLMs when evaluated across a diverse set of languages? RQ3: Is the proposed cosine similarity metric correlated with the proportion of a language in the LLMs' pre-training corpus? RQ4: Is the proposed cosine similarity metric correlated with performance on other benchmark tasks for quantifying the multilingual capabilities of LLMs?

Can Language Ranker Quantify LLM Performance Across Languages? (RQ1)

To visualize the performance of different LLMs in these languages, we selected 10 representative languages to display their inference results. They consist of five high-resource languages, including German, Spanish, French, Indonesian, and Chinese, and five low-resource languages, including Igbo, Kazakh, Kannada, Oriya, and Turkmen. Figure 1 shows detailed results, where the X-axis represents different layers of LLMs, while the Y-axis represents the similarity between the target language and English for each layer. From Figure 1, we can observe that high-resource languages have representations more similar to English, whereas low-resource languages show less similarity. Specifically, German, Spanish, French, and Malay generally maintain cosine similarity scores above 0.6, with Spanish and French often showing the highest scores, indicating that these languages are better represented in the models' embeddings. In contrast, low-resource languages, such as Igbo, Kazakh, Kannada, Oriya, and Turkmen, display significantly lower cosine similarity scores, often below 0.4. These results show the disparities in performance across languages and highlights the utility of the Language Ranker in quantifying these differences robustly.

Comparison Across Different LLMs (RQ2)

In this section, we analyze how various LLMs perform across multiple languages using the Language Ranker. We

[1] Our code link: https://github.com/lizh9885/Language-Ranker/

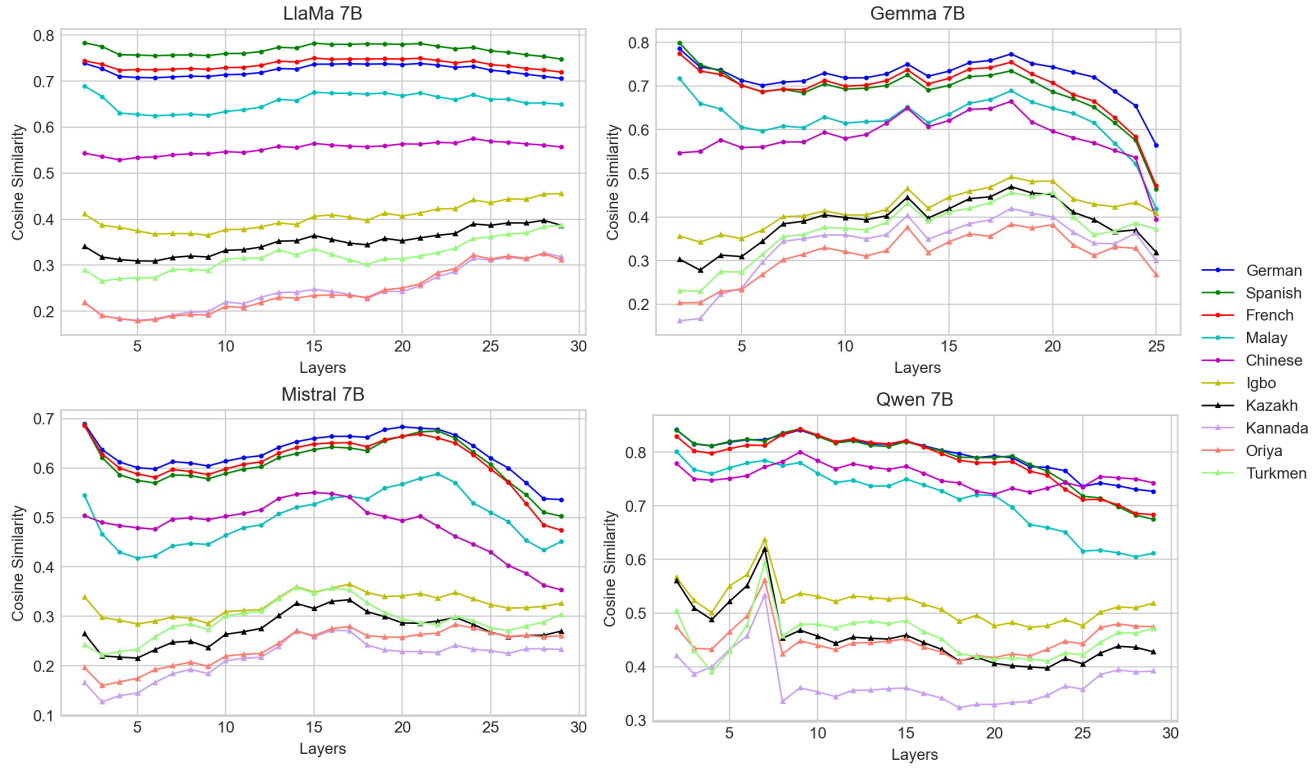

Figure 1: Performance of different LLMs for ten kinds of language. High-resource languages: German, Spanish, French, Indonesian and Chinese; and low-resource languages: Igbo, Kazakh, Kannada, Oriya and Turkmen.

focus on understanding the consistency of language performance among different LLMs, including models with varying architectures and training specifics. We have the following findings:

(1) **Different models display similar results across languages**. Figure 1 presents the cosine similarity scores across various layers for four different 7B parameter LLMs: LlaMa2, Gemma, Mistral, and Qwen. Four LLMs display a similar trend where high-resource languages (i.e., German, Spanish, French, Malay, and Chinese) consistently exhibit higher cosine similarity scores compared to low-resource languages (i.e., Igbo, Kazakh, Kannada, Oriya, and Turkmen). Figure 2 further corroborates these findings by comparing the rank correlation of the similarity scores across the LLMs. Each LLM's ranking is used as a baseline, and the remaining three models exhibit ranking patterns that are largely similar to this baseline. This similarity indicates that despite differences in model architecture or training specifics, the relative performance of languages remains consistent across these four models.

(2) **Fine-tuning on specific languages will improve its performance**. According to the technical report of Qwen (Bai et al. 2023), Qwen has additional fine-tuning on the Chinese corpus, which leads to better performance in Chinese. In Figure 1, we observe that for LlaMa2, Gemma, and Mistral, the performance of Chinese is slightly lower than that of other high-resource languages. However, for Qwen, the performance of Chinese is roughly comparable to other high-resource languages and even shows a gradual improvement in the last few layers. This improvement is more clear in Qwen, mainly due to additional fine-tuning of the Qwen model family on the Chinese corpus, as noted in the technical report of Qwen.

Language	Proportion	Similarity	Language	Proportion	Similarity
German	0.17%	0.723	Welsh	≤0.01%	0.396
French	0.16%	0.737	Persian	≤0.01%	0.300
Swedish	0.15%	0.662	Urdu	≤0.01%	0.275
Chinese	0.13%	0.552	Kannada	≤0.01%	0.236

Table 1: The proportion of different languages in the LlaMa2 pre-training corpus and the similarity metric we proposed. The English language ratio is 89.7%.

(3) **Comparison of Larger LLM Sizes.** We extended our analysis to explore the performance of the similarity metric in larger language models with 13 billion parameters. Figure 3 presents the results for LlaMa2 13B and Qwen 13B. The findings reveal that the general trends observed in 7B models persist in these larger models. Specifically, the relative performance of high-resource and low-resource languages remains largely unchanged, with a clear separation between the two groups. LlaMa2 13B exhibits more pronounced fluctuations in the initial layers compared to its 7B counterpart, suggesting potentially richer early-stage language representations. These observations highlight both the consistency of language performance patterns across model sizes and the potential for larger models to enhance representations for specific languages.